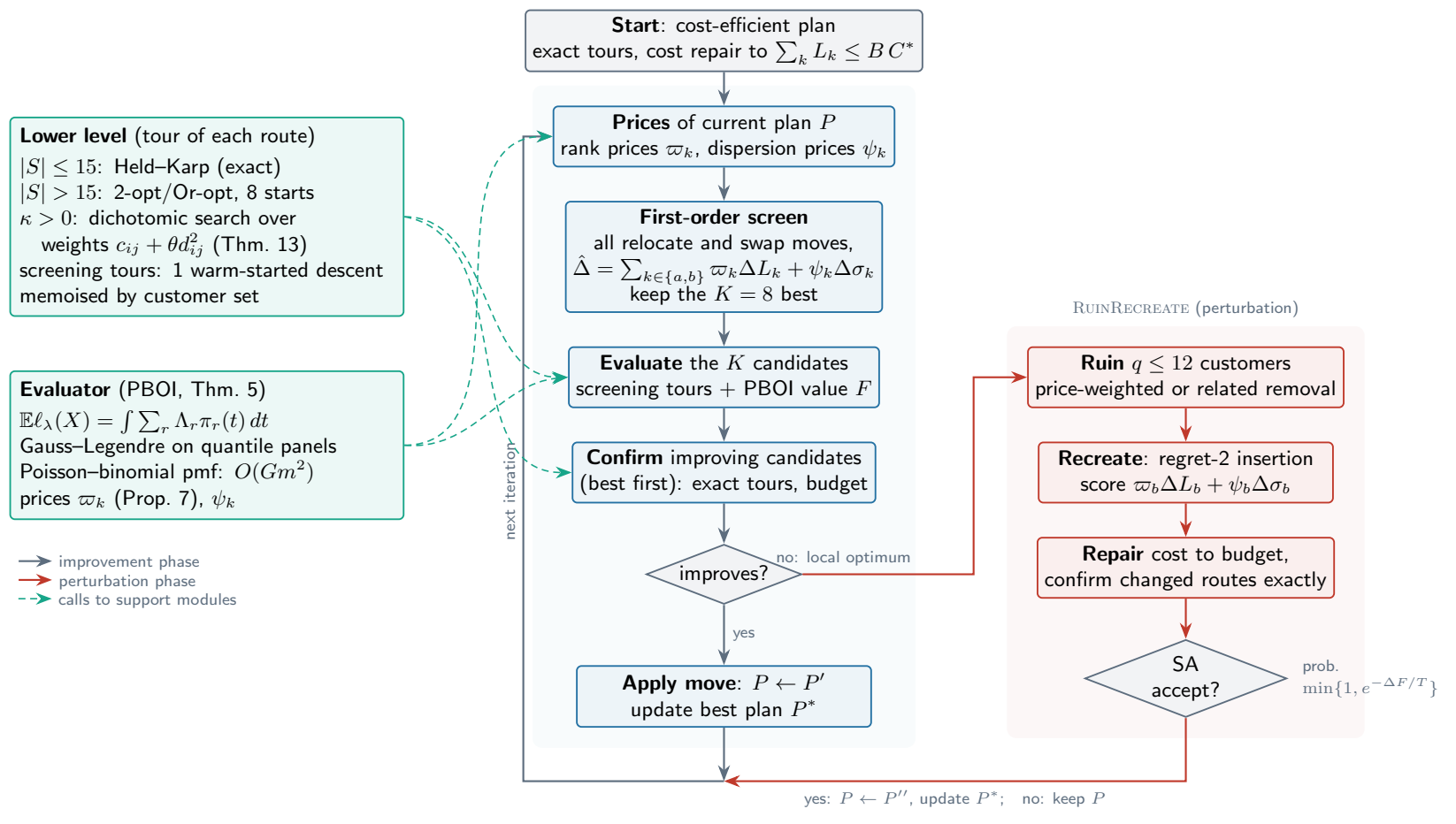

Start: cost-efficient plan
exact tours, cost repair to $\sum_k L_k \le B\,C^*$
Prices of current plan P
rank prices ϖ_k, dispersion prices ψ_k
First-order screen
all relocate and swap moves,
$\hat{\Delta} = \sum_{k\in\{a,b\}} \varpi_k \Delta L_k + \psi_k \Delta\sigma_k$
keep the $K = 8$ best
Evaluate the K candidates
screening tours + PBOI value F
Confirm improving candidates
(best first): exact tours, budget
improves?
no: local optimum
yes
Apply move: $P \leftarrow P'$
update best plan P^*
next iteration
RuinRecreate (perturbation)
Ruin $q \le 12$ customers
price-weighted or related removal
Recreate: regret-2 insertion
score $\varpi_b \Delta L_b + \psi_b \Delta\sigma_b$
Repair cost to budget,
confirm changed routes exactly
SA
accept?
prob.
$\min\{1, e^{-\Delta F/T}\}$
yes: $P \leftarrow P''$, update P^*; no: keep P
Lower level (tour of each route)
$|S| \le 15$: Held–Karp (exact)
$|S| > 15$: 2-opt/Or-opt, 8 starts
$\kappa > 0$: dichotomic search over
weights $c_{ij} + \theta d_{ij}^2$ (Thm. 13)
screening tours: 1 warm-started descent
memoised by customer set
Evaluator (PBOI, Thm. 5)
$\mathbb{E}\ell_\lambda(X) = \int \sum_r \Lambda_r \pi_r(t)\,dt$
Gauss–Legendre on quantile panels
Poisson–binomial pmf: $O(Gm^2)$
prices ϖ_k (Prop. 7), ψ_k
improvement phase
perturbation phase
calls to support modules